Adapting Domain-Aware Knowledge to Vision-Language Model for Zero-Shot Anomaly Detection

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***Abstract*—Zero-shot anomaly detection (ZSAD) is a challenging task that aims to detect anomalies in images without any prior knowledge of the anomaly classes. This task is especially difficult because anomalies are rare, diverse, and often manifest differently across domains, making it hard for models to generalize when training data is scarce or unavailable. Recently, vision-language models (VLMs), such as CLIP, have shown great potential in ZSAD, but they often struggle to adapt to unseen domains due to the lack of domain-aware knowledge. To address these challenges, we propose the Domain Adaptation CLIP (DACLIP), a novel approach that adapts domain-aware knowledge to the VLM. Specifically, DACLIP leverages a Domain-Aware Knowledge Adaptation (DAKA) strategy to enhance CLIP for ZSAD across different domains. The DAKA strategy comprises multiple experts that specialize in target domains, enabling the model to dynamically select and combine specialized experts tailored to anomaly characteristics, thus improving its ability to generalize and detect a wide range of anomalies. Furthermore, we introduce learnable domain-aware prompts that are jointly learned by and injected into both the CLIP encoders (visual and text) and the DAKA modules. This dual-pathway learning enables the model to capture domain-specific features at multiple levels of the architecture, allowing for more effective adaptation to new domains and anomaly types. We evaluate our approach on several benchmark datasets spanning industrial and medical domains. Extensive experiments demonstrate that DACLIP consistently outperforms state-of-the-art methods in ZSAD, achieving significant improvements in both image-level and pixel-level anomaly detection tasks.**

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I. Introduction

ANOMALY detection (AD) holds great significance in various real-world applications [1], [2], [3], [4], [5], such as industrial quality control [6], [7] and medical diagnosis [8], [9]. The primary objective of AD is to identify instances that deviate from the normal distribution, which is challenging due to the scarcity of labeled data. In many scenarios, collecting a large amount of labeled data for training is infeasible; hence, most AD methods [10], [11], [12], [13], [14] utilize unsupervised learning to train models using only normal samples. However, this assumption can be challenged in certain situations, such as: i) when training data access is restricted due to privacy concerns (e.g., safeguarding patient confidentiality), or ii) when the target domain presents entirely new conditions without available training data (e.g., detecting defects in newly introduced manufacturing processes). Zero-shot anomaly detection (ZSAD) aims to address these challenges by enabling the model to detect anomalies in unseen domains without any prior knowledge of the anomaly classes.

Recently, VLMs, such as CLIP, have attracted considerable attention across various applications [15], [16], [17] for their capability to capture rich visual representations from extensive datasets. To utilize VLMs for ZSAD, several methods design or learn text prompts [16], [17] to capture the textual semantics of normal and abnormal patterns, enabling the detection of visual anomalies through semantic matching. WinCLIP [16] is a pioneering work in the ZSAD field, which leverages a diverse set of handcrafted text prompts to harness CLIP's generalization capability for ZSAD. However, WinCLIP demonstrates limited detection performance because its underlying VLM, CLIP [15], is trained on natural image-text datasets [18] and lacks specialization for anomaly detection. VLMs like CLIP are primarily trained to align with the class semantics of foreground objects rather than the concepts of abnormality or normality in images. Consequently, their ability to generalize in recognizing visual abnormality or normality is limited, resulting in suboptimal ZSAD performance.

In the realm of VLM, prompt learning involves incorporating learnable tokens into the input image or text, effectively tailoring VLMs to specific scenarios. For instance, CoOp [19]

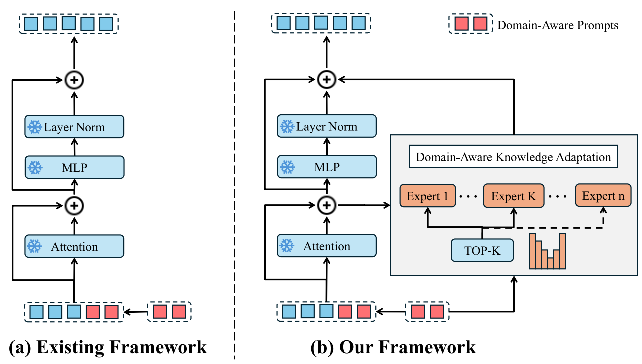

Fig. 1. The comparison of the existing framework and the proposed DACLIP framework for zero-shot anomaly detection.

integrates learnable tokens in addition to the input text into the text branch. Adapting CLIP with learnable text prompts effectively improves its ability to represent normal and abnormal classes, thereby enhancing generalization to unseen domains. Existing works generally adopt the framework presented in Fig. 1(a) to adapt CLIP with prompt learning for ZSAD. AnomalyCLIP [17] is a recent work that introduces learnable text prompts to enhance the performance of CLIP for ZSAD. AdaCLIP [20] introduces hybrid learnable prompts to further improve the performance of CLIP in anomaly detection tasks. However, due to the limited capacity of prompt learning, the text encoder often struggles to produce sufficiently discriminative representations to distinguish subtle anomalies from normal variations, resulting in decreased anomaly detection accuracy. These methods rely heavily on prompt learning, which often fails to capture the nuanced domain-aware knowledge required for effective anomaly detection. Overall, these methods often fail to incorporate fine-grained and domain-aware considerations, resulting in a decline in performance.

To tackle these challenges, we propose a novel approach called DACLIP, which leverages the DAKA strategy to adapt CLIP for ZSAD across different domains. The overview of adapting VLM in ZSAD is illustrated in Fig. 2. The main difference between the existing framework and our proposed DACLIP framework is shown in Fig. 1. The DAKA strategy is derived from the Mixture-of-Experts (MoE) framework [21], [22] and is particularly well-suited for ZSAD. ZSAD faces a fundamental challenge: a single universal model must generalize across diverse domains with distinct visual characteristics and anomaly semantics. A monolithic CLIP model produces fixed embeddings regardless of domain context, leading to representation collapse, where the model cannot simultaneously optimize for conflicting requirements across domains. DAKA strategy addresses this through conditional computation: each expert specializes by receiving gradients only when selected by the gating network. The domain-aware gating mechanism selects relevant experts based on input and domain context, addressing ZSAD's core challenges of domain diversity and anomaly heterogeneity. To further enhance domain-specific adaptation, we introduce learnable domain-aware prompts that are jointly optimized with both CLIP encoders and DAKA modules during training on auxiliary domain data. The prompts serve as a bridge between CLIP and DAKA: they adapt CLIP features via residual corrections while simultaneously guiding expert selection through the gating mechanism (Eq. (5)). This dual-pathway learning enables the model to capture domain-specific features at multiple architectural levels, allowing for more effective adaptation to new domains and anomaly types. In contrast to previous methods that rely solely on text prompts or generic learnable prompts, our approach combines the strengths of both text and visual prompts injected into both CLIP and DAKA modules. This multi-level, dual-modality adaptation ensures that domain-aware knowledge is propagated throughout the vision-language framework, leading to more robust and generalizable anomaly detection across diverse domains. In summary, we make the following contributions:

- This paper proposes a ZSAD method named DACLIP, which leverages the DAKA strategy to adapt domain-aware knowledge into VLM for ZSAD. This paper introduces the DAKA strategy that consists of multiple experts specializing in target domains, allowing the model to learn domain-aware features and improve its performance on unseen domains.
- To further exploit the domain information, this paper introduces learnable domain-aware visual and text prompts. The key difference and innovation of our approach is the injection of learnable domain-aware prompts into both CLIP and the DAKA modules. This enables the model to effectively capture domain-specific features throughout the vision-language framework, thereby enhancing its adaptability and performance on unseen domains.
- We conduct extensive experiments on several benchmark datasets, including industrial and medical datasets. Our approach achieves state-of-the-art performance in ZSAD, demonstrating its effectiveness in both image-level and pixel-level anomaly detection tasks.

II. Related Work

A. Conventional Anomaly Detection

Various conventional anomaly detection methods have been proposed based on unsupervised learning. Unsupervised methods mainly rely on the normal samples, and they aim to learn a model that can distinguish between normal and abnormal samples. These methods can be categorized into three main groups: Synthesis-based methods, Embedding-based methods, and Reconstruction-based methods. Synthesis-based methods [23], [24], [25], [26] generate artificial anomalies on normal images and utilize both the original normal images and the synthesized anomalies to train the model. For example, DRAEM [23] introduces a method to simulate out-of-distribution anomalous samples and trains both a reconstructive subnetwork and a discriminative subnetwork. Embedding-based methods [11], [12], [27], [28] utilize pre-trained networks to extract feature representations, enabling the detection and localization of anomalies. Approaches like PatchCore [12] construct a memory bank of normal embeddings and identify anomalies by measuring the distance between a test sample and its closest normal embedding. Reconstruction-based methods [10], [29], [30], [31], [32], [33], [34] focus on training models to recreate normal images, where anomalies are detected by analyzing elevated reconstruction errors. UniAD [30] employs a transformer network equipped with a masked self-attention mechanism to reconstruct features, ensuring the model avoids collapsing into an identity mapping.

B. Zero-Shot Anomaly Detection

ZSAD has become feasible with the advancement of large pre-trained foundation models, particularly VLMs. CLIP [15] has been extensively utilized as a VLM to facilitate zero-shot anomaly detection (ZSAD) on visual data [16], [17], [35], [36]. CLIP, however, faces challenges with insufficient domain-aware sensitivity, which restrict its effectiveness in detecting anomalies. To tackle this issue, early approaches relied on manually crafted prompt templates [16], [37], which heavily depend on domain expertise and the quality of the prompts. WinCLIP [16] represents an early effort in the ZSAD domain, utilizing a variety of manually designed text prompts to tap into CLIP's potential for generalization in anomaly detection. However, handcrafted prompts often lead to suboptimal performance, as they may not effectively capture the intricate semantics of anomalies. To address this limitation, APRIL-GAN [38] employs learnable linear projection techniques to improve the representation of local visual semantics. SDP [39] utilizes multi-level features and incorporates architectural and feature modifications to enhance its performance. More recent works have adopted prompt learning techniques [17], [20], [40], [41] to automate prompt optimization. AnomalyCLIP [17] introduces object-agnostic prompts, streamlining the prompt design process by leveraging general anomalous patterns. However, these methods often struggle to capture the domain-aware features necessary for effective anomaly detection, leading to subpar performance. This paper leverages the DAKA strategy to enhance CLIP for ZSAD across different domains. The DAKA strategy is introduced to specialize in target domains, allowing the model to learn domain-aware features and improve its performance on unseen domains. The learnable domain-aware prompts are learned by both CLIP and the DAKA, enabling the model to capture domain-aware features and enhance its performance on unseen domains.

C. Mixture-of-Experts

Mixture-of-Experts (MoE) models have gained significant attention in recent years due to their ability to improve model performance while maintaining efficiency. The MoE approach has been extensively explored in fields such as computer vision [42], [43], natural language processing [44], and vision-language pretraining [45]. This method leverages a collection of specialized experts to enhance model scalability, with a gating mechanism dynamically allocating the contribution of each expert based on the input. MoE is well-suited for multimodal learning; however, the inherent differences between modalities pose significant challenges. LIMoE incorporates additional auxiliary losses to better balance and align different modalities [46]. MoSA [47] employs a mixture of sparse adapters to enhance the performance of large language models (LLMs) in various tasks. Although MoE demonstrates significant potential, its utilization in transfer learning for cross-modal downstream tasks has received limited attention and remains underdeveloped. In this paper, we propose a DAKA strategy that effectively integrates with CLIP to enhance its performance in ZSAD tasks.

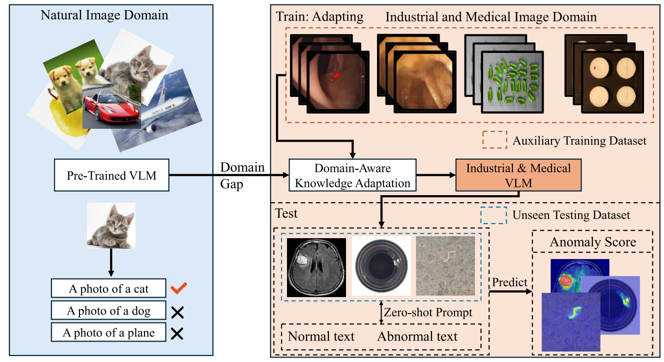

Fig. 2. The overview of the Domain-aware Knowledge Adaptation in pre-trained VLM for zero-shot anomaly detection.

III. Method

A. Problem Statement

In the context of ZSAD [17], [48], [49], our approach leverages an auxiliary anomaly detection dataset to train the model, enabling it to generalize effectively to unseen domains. This paper denotes $\mathcal{D}_{train} = \{\mathcal{X}_{train}, \mathcal{Y}_{train}\}$ as an auxiliary training dataset consisting of both normal and anomalous samples. $\mathcal{X}_{train} = \{x_i\}_{i=1}^{N}$ is a set of N iamges. $\mathcal{Y}_{train} = \{y_i, G_i\}_{i=1}^{N}$ are the corresponding set of truth labels and pixel-level anomaly masks. Each image x_i is associated with a label $y_i \in \{0, 1\}$, where 0 indicates a normal sample and 1 indicates an anomalous sample. The anomaly masks G_i are binary masks that indicate the presence of anomalies in the images. During the testing phase, the model is evaluated on a collection of unseen test datasets. $\mathcal{D}_{test} = \{\mathcal{X}_{test}, \mathcal{Y}_{test}\}$ is a test dataset and $\mathcal{X}_{test} = \{x_i\}_{i=1}^{M}$ is a set of M images and $\mathcal{Y}_{test} = \{y_i, G_i\}_{i=1}^{M}$ are the corresponding set of truth labels and pixel-level anomaly masks. The objective is to develop a model that associates an input image x_i with its an image-level anomaly score and a pixel-level anomaly map.

B. Overview of DACLIP

We propose the DACLIP, a method that leverages CLIP as a zero-shot backbone for anomaly detection by addressing the domain gap between CLIP's pretraining and the requirements of specialized anomaly detection tasks. The overall architecture of DACLIP is illustrated in Fig. 3. The proposed DACLIP method consists of three main components: a CLIP model, DAKA modules, and learnable domain-aware prompt tokens. Given an image, DACLIP adheres to the general ZSAD principle by comparing CLIP embeddings, similar to the approach employed by WinCLIP [16]. Anomalies can be detected by computing similarities in the CLIP embedding

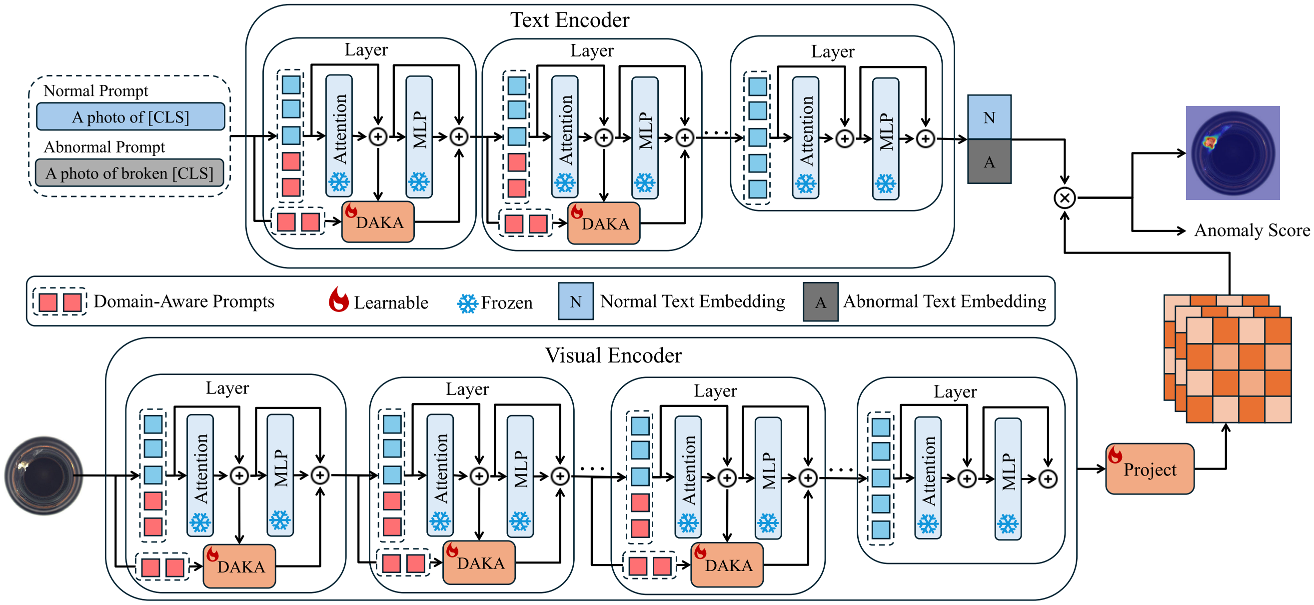

Fig. 3. Overview of the proposed DACLIP, which consists of a CLIP model, DAKA modules and prompt tokens. The DAKA module is composed of multiple experts, specializing in the target domain. The learnable domain-aware prompts are learned by both the CLIP and the DAKA module, allowing the model to capture domain-aware features and improve its performance on unseen domains. The domain-aware prompts are learned from the input images and text, and they are used to guide the model in focusing on the relevant features for each domain.

space between the image and textual captions representing normal and abnormal states, such as "A photo of a normal [CLS]" and "A photo of an abnormal [CLS]", where [CLS] denotes to the name of the testing category. DACLIP learns domain-aware prompts for both the image and text encoders. Notably, these prompts are learned by both the CLIP and the DAKA modules, allowing the model to capture domain-aware features and improve its performance on unseen domains.

C. Domain-Aware Knowledge Adaptation

1) Prompt Learning: Commonly used text prompt templates in CLIP, such as "A photo of a [CLS]," are primarily designed to capture object-level semantics. However, these templates may not effectively represent the concepts of normality and abnormality, leading to suboptimal performance in ZSAD tasks. To facilitate the learning of anomaly-discriminative textual embeddings, we incorporate prior anomaly semantics into the design of text prompt templates. For this purpose, we can adopt the text "A photo of a damaged [CLS]" to encompass comprehensive anomaly semantics, enabling the detection of diverse defects such as scratches and holes. To enhance the learning of more discriminative visual and textual spaces, we draw inspiration from [17], [50] by introducing randomly initialized learnable token embeddings into the text and visual encoders. This approach refines the visual and textual spaces, enabling better adaptation to anomaly detection tasks. To control the degree of refinement in the visual and textual spaces, we substitute the prefix portion of the original token embeddings with learnable token embeddings in both the visual and text encoders. More specifically, the prompt tokens $p = \{p_i^1, p_i^2, \ldots, p_i^{n_p}\} \in R^{n_p \times c}$ are introduced to replace a portion of the input vanilla tokens in the transformer layer, enabling the model to incorporate additional domain-aware contextual information. n_p denotes the length of prompt tokens, i is the index of transformer layer, and c is the dimension of the token embeddings. In particular, we denote the vanilla tokens of the transformer as $T_j = \{t_j^1, t_j^2, \ldots, t_j^{n_t}\} \in R^{n_t \times c}$, where n_t is the number of tokens in the input image, $n_t > n_p$, and j is the index of the transformer layer. To adapt the CLIP model with specific domain knowledge, we replace the first n_p tokens of the vanilla tokens, thus deriving the new token embeddings as follows:

$$T'_j = \{p_j^1, p_j^2, \ldots, p_j^{n_p}, t_j^{n_p+1}, \ldots, t_j^{n_t}\} \in R^{n_t \times c} \tag{1}$$

Learnable prompt tokens are incorporated up to a predefined depth, while the prompt tokens for the remaining layers are generated through a feed-forward mechanism. Thanks to the self-attention mechanism in transformer layers, the learnable prompt tokens influence all output tokens, including the vanilla ones. To further enhance the model's ability to learn domain-aware features, we input the learnable prompt tokens into DAKA modules, as calculated in Eq. (5).

2) Expert Module: DAKA inserts small modules into transformer layers, allowing for efficient adaptation to new tasks without the need for extensive retraining of the entire model. The Expert Module is a lightweight module that can be easily integrated into DAKA. The module comprises a layer normalization, a linear transformation, and a non-linear activation function, such as ReLU. The Expert Module (EM) is formulated as follows:

$$\mathrm{EM}(T') = \mathrm{ReLU}(\mathrm{LN}(T') \cdot W_e) \tag{2}$$

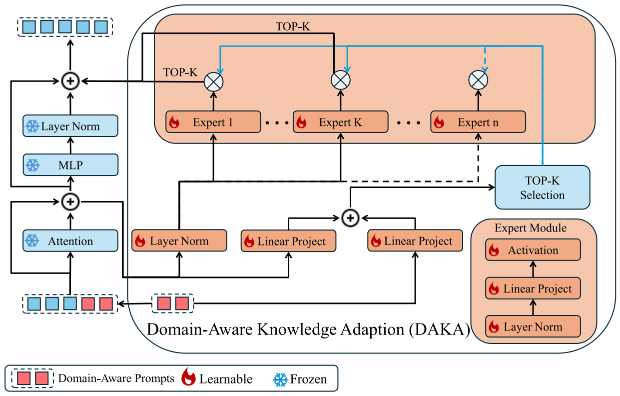

Fig. 4. Overview of the proposed DAKA strategy. The DAKA consists of multiple experts, specializing in a specific domain. Each expert is a lightweight module that can be easily integrated into existing models. The experts are trained to learn domain-aware features, allowing the model to adapt to different domains.

where T' is the input feature tokens defined in Eq. (1), LN denotes layer normalization, and W_e is the learnable weight matrix. The specialization of experts in DAKA is achieved through three complementary mechanisms that operate with the existing adapter and gating modules: (1) A learned routing/gating network conditions on input features and the learnable domain-aware prompts and selects the Top-K experts, as calculated in Eq. (5). (2) All experts share the same lightweight adapter architecture but maintain separate parameter sets; gradients are routed to an expert only when it is selected by the gating network, so each expert naturally specializes on the inputs it serves. (3) Domain-aware prompts supply explicit domain context to both the gating network and the experts, steering routing decisions and expert transformations toward domain-relevant patterns.

3) DAKA Module: As illustrated in Fig. 4, the DAKA module comprises multiple experts and operates in parallel with the MLP layers within each transformer. This design leverages the non-linear transformations of MLP layers to enhance the model's adaptability by incorporating a DAKA module mechanism. The output of the DAKA modules for a given input x is calculated as a weighted combination of the outputs from the selected modules, with the selection determined by the gating network $\mathcal{G}$. To enable fine-grained representations, given n Expert modules, the forward process through the MLP layer can be expressed as:

$$\mathcal{F}_{DAKA}(T') = \sum_{i=1}^{n} \mathcal{G}_i(T') \cdot \mathrm{EM}_i(T') \tag{3}$$

where $\mathcal{G}_i(T')$ is the gating function that determines the weight of the i-th Expert module, and $\mathrm{EM}_i(T')$ is the output of the i-th Expert module. The gating function is a softmax function that computes the weights based on the input tokens T' and the parameters of the gating network. We employ a straightforward yet effective gating mechanism [22], which applies a softmax function over the TopK logits produced by a linear layer.

$$\mathcal{G}(T') = \mathrm{Softmax}(\mathrm{TopK}(T' \cdot W_g)) \tag{4}$$

where TopK refers to selecting the highest K weights, and W_g is the learnable weight matrix for the gating network. However, this routing mechanism overlooks domain-aware information, limiting its effectiveness when adapting foundation models to specialized domain tasks. To address this limitation, we introduce a domain-aware gating mechanism that incorporates domain-aware information into the routing process. The domain-aware gating mechanism is formulated as follows:

$$\mathcal{G}^p(T') = \mathrm{Softmax}(\mathrm{TopK}(T' \cdot W_g + p \cdot W_p)) \tag{5}$$

where p is the domain-aware prompt tokens, which is also learned in the transformer layers (Eq. 1), and W_p is the learnable weight matrix for the domain-aware gating network. Unlike static prompts, these domain-aware prompts are jointly optimized with the DAKA modules and transformer layers during training on auxiliary domain data through the primary anomaly detection objective. Through this optimization, the DACLIP learns to encode domain-specific semantic and visual patterns that are most discriminative for distinguishing normal and anomalous samples in each domain. The output after an MLP layer and DAKA modules can be expressed as:

$$\begin{aligned} y_{mlp} &= \mathcal{F}_{MLP}(T') + T' \\ y_{out} &= y_{mlp} + \sum_{i=1}^{n} \mathcal{G}_i^p(T') \cdot \mathrm{EM}_i(T') \end{aligned} \tag{6}$$

where y_{mlp} is the output of the MLP layer, and y_{out} is the final output after summing the outputs of the MLP layer and the DAKA modules.

D. Dimension-Align Projection Layer

The image encoder processes the input image through multiple transformer layers to extract patch embeddings, represented as $\mathbf{F^I} = \{\mathbf{F^I}_0, \mathbf{F^I}_1, \ldots\}$. Simultaneously, the text encoder generates embeddings for the corresponding textual captions, producing normal embeddings $\mathbf{F^T_N}$ and abnormal embeddings $\mathbf{F^T_A}$. These embeddings serve as the foundation for aligning visual and textual representations, enabling effective anomaly detection.

The original CLIP architecture exhibits a mismatch in dimensions between the patch embeddings generated by the image encoder and the text embeddings produced by the text encoder. To address this issue, we introduce a projection layer, which is appended to the image encoder. The primary function of the projection layer is to align the dimensions of the patch embeddings ($\mathbf{F^I}$) with the text embeddings of normal ($\mathbf{F^T_N}$) and abnormal ($\mathbf{F^T_A}$) textual captions. This alignment is achieved by employing a linear transformation with a bias term. The projection layer applies a learnable linear mapping to the patch embeddings, ensuring compatibility with the text embeddings.

E. Training and Inference

1) Anomaly Detection and Localization: We calculate the anomaly score by evaluating the similarities between the patch embeddings $\mathbf{F^I}$ and the text embeddings $\mathbf{F^T_N}$ (normal) and $\mathbf{F^T_A}$ (abnormal).

$$S_x^A(i,j) = \frac{\exp(\mathbf{F^I}(i,j)\mathbf{F^T_A}^\top)}{\exp(\mathbf{F^I}(i,j)\mathbf{F^T_A}^\top) + \exp(\mathbf{F^I}(i,j)\mathbf{F^T_N}^\top)}, \tag{7}$$

where $[\cdot]^{\top}$ denotes the transpose operation, $\mathbf{F}^{\mathbf{I}}(i, j)$ represents the token embedding of the patch centered at (i, j), and $S_x^A(i, j)$ is the anomaly score at the patch level. The corresponding normal scores can be calculated via $S_x^N(i, j)$ using the similarity to F_N^T. We adopt a multi-hierarchy approach by extracting anomaly maps from several layers and aggregating these maps to produce the final prediction. This strategy leverages information from multiple levels of representation, enhancing the accuracy and robustness of anomaly detection.

2) Loss Function: The image-level anomaly score $\mathcal{S}(x_i)$ is calculated similar to Eq. (7) and then optimized by minimizing the following loss on the training set:

$$\mathcal{L}_{global} = \frac{1}{N}\sum_{i=1}^{N}\mathcal{L}_{focal}(\mathcal{S}(x_i), y_i) \tag{8}$$

where N represents the total number of training samples, and the focal loss [51] is utilized to optimize the image-level anomaly score. DACLIP generates abnormality-oriented segmentation map $\mathcal{M}^{\mathcal{A}} \in R^{h\times w}$, where each entry is computed using $S_x^A(i, j)$ as defined in Eq. (7):

$$\mathcal{M}^{\mathcal{A}} = \phi(S_x^A) \tag{9}$$

where ϕ is a reshape and interpolation function that converts the patch-level anomaly scores into a segmentation map. Similarly, the segmentation map for normality $\mathcal{M}^{\mathcal{N}}$ can be generated by utilizing the normal scores $S_x^N(i, j)$, which are computed based on the similarity to the normal text embeddings. This paper employs the Dice loss [52] and Focal loss [51] to optimize the pixel-level segmentation maps, which is defined as follows:

$$\begin{aligned}\mathcal{L}_{local} = \frac{1}{N}\sum_{i=1}^{N}&\mathcal{L}_{focal}([\mathcal{M}^{\mathcal{A}}(x_i), \mathcal{M}^{\mathcal{N}}(x_i)], G_i)\\ &+ \mathcal{L}_{Dice}(\mathcal{M}^{\mathcal{A}}(x_i), G_i)\\ &+ \mathcal{L}_{Dice}(\mathcal{M}^{\mathcal{N}}(x_i), 1 - G_i)\end{aligned} \tag{10}$$

Overall, DACLIP is optimized by minimizing the following combined loss, which integrates both local and global objectives:

$$\mathcal{L} = \mathcal{L}_{global} + \mathcal{L}_{local} \tag{11}$$

IV. Experiments

A. Experimental Setup

1) Datasets: We conducted experiments on datasets from both industrial and medical domains. Specifically, for the industrial domain, we use MVTec-AD [6], VisA [7], MPDD [53], BTAD [54], KSDD [55], DAGM [56], and DTD-Synthetic [57]. For the medical domain, we utilize HeadCT [58], BrainMRI [59], Br35H [60], ISIC [61], CVC-ColonDB [62], CVC-ClinicDB [63], Kvasir [64], Endo [65], and TN3K [66]. The statistics of the datasets are summarized in Table III.

2) Evaluation Metrics: To align with prior ZSAD studies [16], [17], [49], we utilize the Area Under the Receiver Operating Characteristic Curve (I-Auroc and P-Auroc), Average Precision (I-AP and P-AP), and F1 score (I-F1 and P-F1) at the optimal threshold to assess performance at both the image-level and pixel-level.

3) Implementation Details: In this study, we utilize the pre-trained CLIP (ViT-L/14@336px) model as the default backbone. Patch embeddings are extracted from the 6th, 12th, 18th, and 24th transformer layers to capture multi-level feature representations. We resize all input images to 518×518 resolution during both training and inference to ensure consistency in input dimensions. This resolution choice is consistent with recent state-of-the-art ZSAD methods such as AnomalyCLIP and APRIL-GAN, ensuring fair experimental comparison. We set the number of epochs to 5 for training on auxiliary datasets, with a batch size of 1. The AdamW optimizer is employed with an initial learning rate of 0.005. By default, we use the industrial dataset MVTec-AD [6] and the medical dataset ClinicDB [63] as auxiliary data for training. For evaluations on MVTec-AD and ClinicDB, we utilize VisA [7] and ColonDB [62] as training datasets. All experiments were conducted using PyTorch-2.0.0 on a single NVIDIA RTX 3090 GPU with 24GB of memory.

B. Comparison With State-of-the-Arts

1) Compared Methods: This study compares the performance of DACLIP with several state-of-the-art methods, including SDP [39], WinCLIP [16], APRIL-GAN [38], and AnomalyCLIP [17]. WinCLIP introduces a method that divides images into windows and computes a separate class token for each window to represent its corresponding position. This approach generates a feature map composed of class tokens, which can then be compared with text features to produce the anomaly map. SDP introduces a Staged Dual-Path model that utilizes features from multiple levels and incorporates architectural and feature modifications to enhance anomaly detection performance. APRIL-GAN proposes a solution based on the CLIP model by incorporating additional linear layers to enhance the representation of local visual semantics. AnomalyCLIP learns object-agnostic text prompts that capture generic patterns of normality and abnormality in an image, independent of its foreground objects.

2) Zero-Shot Anomaly Detection in the Industrial Domain: Table I presents the performance of DACLIP and other methods on the industrial domain related datasets. DACLIP demonstrates superior ZSAD performance across the datasets, significantly outperforming the other four methods on the majority of the datasets. Compared to the second-best method (AnomalyCLIP), DACLIP achieves an average improvement of 1.3% in I-Auroc and 0.6% in I-AP on image-level. On pixel-level, DACLIP achieves an average improvement of 5.8% in P-AP and 3.4% in P-F1 compared to the AnomalyCLIP. AnomalyCLIP achieves the better performance on the I-F1 and P-Auroc metrics. Compared to SDP, DACLIP achieves an average improvement of 5.8% in I-Auroc, 3.0% in I-AP and 1.4% in I-F1 on image-level. On the pixel level, DACLIP demonstrates significant improvements compared to SDP. Although WinCLIP and VAND achieve excellent results by using manually defined text prompts, their performance is still inferior to DACLIP. Furthermore, DACLIP underperforms on some datasets, such as MVTec-AD and BTAD, when compared to AnomalyCLIP. This discrepancy

TABLE I

ZSAD Performance Comparison on Industrial Domain. The Best Performance Is Highlighted in Bold, and the Second-Best Is Underlined

Metrics	Datasets	SDP	WinCLIP	APRIL-GAN	AnomalyCLIP	DACLIP
I-Auroc, I-AP, I-F1	MVTec-AD	(90.9, 95.8, 91.9)	(**91.8**, **96.5**, **92.9**)	(86.0, 93.6, 90.4)	(91.5, 96.2, 92.7)	(90.7, 95.8, 92.0)
	VisA	(78.6, 81.5, 79.2)	(78.1, 81.2, 79.0)	(78.0, 80.8, 78.7)	(80.7, 84.4, 79.9)	(**85.2**, **87.8**, **82.4**)
	MPDD	(59.8, 69.4, 78.1)	(61.5, 69.2, 77.5)	(76.6, **82.8**, **81.9**)	(75.0, 76.9, 80.2)	(**80.1**, **82.8**, 81.6)
	BTAD	(84.4, 87.5, 82.2)	(68.2, 70.9, 67.8)	(74.2, 70.1, 68.7)	(88.6, **90.6**, **86.2**)	(**90.5**, 82.0, 78.0)
	KSDD	(94.0, 85.9, 81.6)	(93.4, 86.2, 78.9)	(**96.8**, **92.3**, **88.9**)	(95.1, 82.5, 75.9)	(96.0, 89.5, 83.0)
	DAGM	(90.3, 91.5, 86.2)	(91.7, 92.7, 87.7)	(94.8, 94.8, 91.1)	(**98.0**, **97.6**, **96.1**)	(95.5, 95.1, 92.8)
	DTD-Synthetic	(93.0, 97.3, 93.4)	(**95.1**, **97.7**, 94.1)	(85.5, 93.9, 89.0)	(93.7, 97.5, **94.4**)	(93.1, 97.0, 92.6)
	Average	(84.4, 87.0, 84.7)	(82.8, 84.9, 82.6)	(84.6, 86.9, 84.1)	(88.9, 89.4, **86.5**)	(**90.2**, **90.0**, 86.1)
P-Auroc, P-AP, P-F1	MVTec-AD	(87.5, 30.4, 35.9)	(85.1, 18.2, 31.7)	(87.6, **40.7**, 43.2)	(**91.0**, 34.3, 38.8)	(88.0, 39.1, 42.2)
	VisA	(88.1, 12.2, 17.0)	(79.6, 5.4, 14.8)	(94.2, 25.8, **32.3**)	(**95.6**, 21.5, 28.3)	(**95.6**, **26.3**, **32.3**)
	MPDD	(63.6, 14.7, 16.6)	(71.2, 14.1, 15.4)	(95.2, 24.7, 29.6)	(**96.0**, **27.6**, **32.7**)	(**96.4**, 26.5, 30.9)
	BTAD	(90.5, 33.7, 37.4)	(72.7, 12.9, 18.5)	(91.4, 32.4, 37.4)	(92.7, 40.8, 46.6)	(92.2, **44.7**, **47.9**)
	KSDD	(97.2, 23.1, 35.1)	(95.8, 11.7, 21.3)	(94.1, 14.8, 26.0)	(**97.9**, **41.2**, **51.0**)	(97.1, 40.8, 48.2)
	DAGM	(85.5, 26.9, 32.3)	(81.3, 8.6, 13.9)	(89.8, 47.4, 50.1)	(**96.4**, 58.4, 59.9)	(94.3, **62.0**, **62.6**)
	DTD-Synthetic	(91.8, 37.7, 42.6)	(79.5, 9.8, 16.1)	(96.5, 67.7, 65.5)	(97.5, 52.6, 55.8)	(**98.6**, **77.9**, **72.6**)
	Average	(86.3, 25.5, 31.0)	(80.7, 11.5, 18.8)	(92.7, 36.2, 40.6)	(**95.3**, 39.5, 44.7)	(94.6, **45.3**, **48.1**)

TABLE II

ZSAD Performance Comparison on Medical Domain.The Best Performance Is Highlighted in Bold, and the Second-Best Is Underlined.Note That the Image-Level Medical AD Datasets Do Not Contain Segmentation Ground Truth, So the Pixel-Level Medical AD Datasets Are Different From the Image-Level Datasets

Metrics	Datasets	SDP	WinCLIP	APRIL-GAN	AnomalyCLIP	DACLIP
I-Auroc, I-AP, I-F1	HeadCT	(91.0, **92.7**, 84.8)	(84.1, 84.1, 79.8)	(86.9, 87.8, 81.6)	(**93.5**, 91.8, **89.6**)	(91.4, 90.9, 84.6)
	BrainMRI	(89.7, 93.8, 85.2)	(89.9, 93.5, 86.6)	(93.0, 94.2, 91.5)	(91.6, 93.6, 89.2)	(**95.6**, **97.1**, **92.4**)
	Br35H	(87.1, 87.8, 79.2)	(81.6, 84.0, 74.4)	(93.1, 93.8, 85.7)	(95.3, 95.2, 88.3)	(**96.4**, **96.3**, **91.0**)
	Average	(89.3, 91.4, 83.1)	(85.2, 87.2, 80.3)	(91.0, 92.0, 86.3)	(93.5, 93.5, 89.0)	(**94.5**, **94.8**, **89.3**)
P-Auroc, P-AP, P-F1	ISIC	(88.8, 74.0, 70.4)	(67.1, 44.8, 48.5)	(85.0, 69.1, 68.6)	(89.7, 76.6, 71.8)	(**92.3**, **84.3**, **79.0**)
	CVC-ColonDB	(76.6, 21.2, 27.3)	(61.2, 13.3, 19.6)	(78.1, 23.1, 32.2)	(83.0, 32.4, 38.0)	(**89.5**, **59.2**, **55.6**)
	CVC-ClinicDB	(71.5, 16.0, 26.3)	(67.1, 15.4, 24.4)	(78.5, 30.6, 38.4)	(82.2, 35.0, 41.6)	(**90.1**, **59.5**, **56.5**)
	Kvasir	(64.6, 22.7, 32.7)	(38.6, 12.3, 27.0)	(80.1, 43.0, 48.0)	(80.5, 40.7, 48.0)	(**92.8**, **76.1**, **70.9**)
	Endo	(75.3, 35.4, 39.7)	(43.7, 12.8, 25.3)	(83.5, 47.4, 50.6)	(85.3, 47.6, 51.6)	(**86.9**, **57.9**, **57.5**)
	TN3K	(72.4, 28.2, 35.0)	(67.2, 20.5, 30.0)	(72.9, 32.2, 39.0)	(81.0, 44.9, **47.1**)	(**82.7**, **45.4**, 46.3)
	Average	(74.9, 32.9, 38.6)	(57.5, 19.9, 29.1)	(79.7, 40.9, 46.1)	(83.6, 46.2, 49.7)	(**89.1**, **63.7**, **61.0**)

may stem from the fact that AnomalyCLIP employs object-agnostic text prompts, which are particularly effective in datasets with diverse categories. In contrast, DACLIP utilizes domain-aware prompts that are optimized for specific domains, which may not generalize as effectively across all categories. The results indicate that DACLIP is capable of effectively learning domain-aware knowledge and adapting to the target domain, leading to improved performance in ZSAD tasks.

Fig. 5 presents a qualitative comparison of pixel-level anomaly localization performance across five methods on industrial datasets. Notably, DACLIP demonstrates sharper localization of anomalous regions compared to alternatives, particularly in complex textures (e.g., Blotchy). On multiple small objectives (e.g., Tubes), DACLIP effectively localizes defects with minimal deviation from the ground truth, whereas competing methods exhibit over-segmentation or miss fine details. This visualization underscores DACLIP technical advancement in reducing false positives while preserving fine-grained defect boundaries.

TABLE III

Key Statistics on the Datasets Used. $|\mathcal{C}|$ Is the Number of Categories

Domain	Dataset	Category	Modalities	$\lvert\mathcal{C}\rvert$	Normal and anomalous samples
Industrial	MVTec-AD	Obj &texture	Photography	15	(467, 1258)
	VisA	Obj	Photography	12	(962, 1200)
	MPDD	Obj	Photography	6	(176, 282)
	BTAD	Obj	Photography	3	(451, 290)
	KSDD	Obj	Photography	1	(181, 74)
	DAGM	Texture	Photography	10	(6996, 1054)
	DTD-Synthetic	Texture	Photography	12	(357, 947)
Medical	ISIC	Skin	Photography	1	(0, 379)
	CVC-ClinicDB	Colon	Endoscopy	1	(0, 612)
	CVC-ColonDB	Colon	Endoscopy	1	(0, 380)
	Kvasir	Colon	Endoscopy	1	(0, 1000)
	Endo	Colon	Endoscopy	1	(0, 200)
	TN3K	Thyroid	Radiology (Utralsound)	1	(0, 614)
	HeadCT	Brain	Radiology (CT)	1	(100, 100)
	BrainMRI	Brain	Radiology (MRI)	1	(98, 155)
	Br35H	Brain	Radiology (MRI)	1	(1500, 1500)

3) Zero-Shot Anomaly Detection in the Medical Domain: Table II presents the performance of DACLIP and other

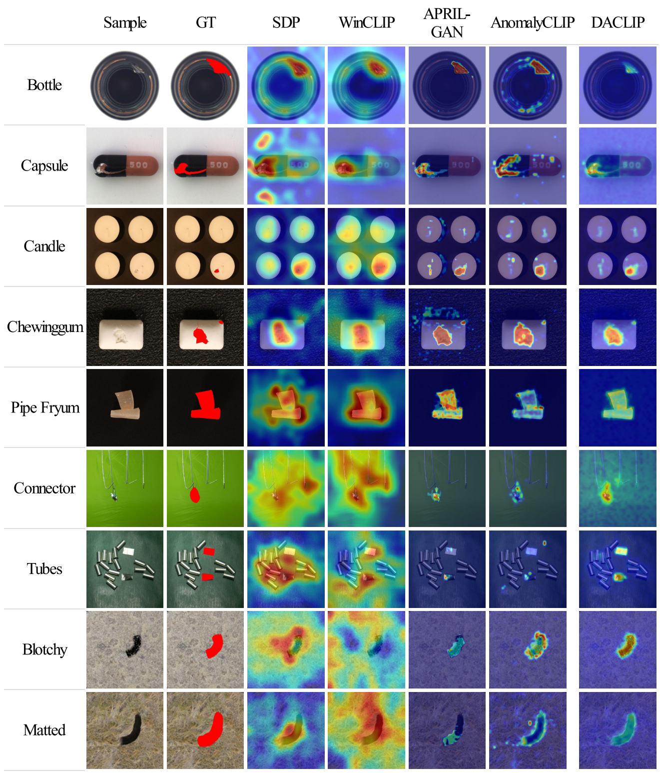

Fig. 5. Qualitative visualization for pixel-level anomaly localization on industrial datasets. The leftmost column lists categories, followed by ground truth (GT) masks with annotated anomalies (red). Subsequent columns visualize detection heatmaps from five methods, where warmer hues denote higher confidence in anomaly detection.

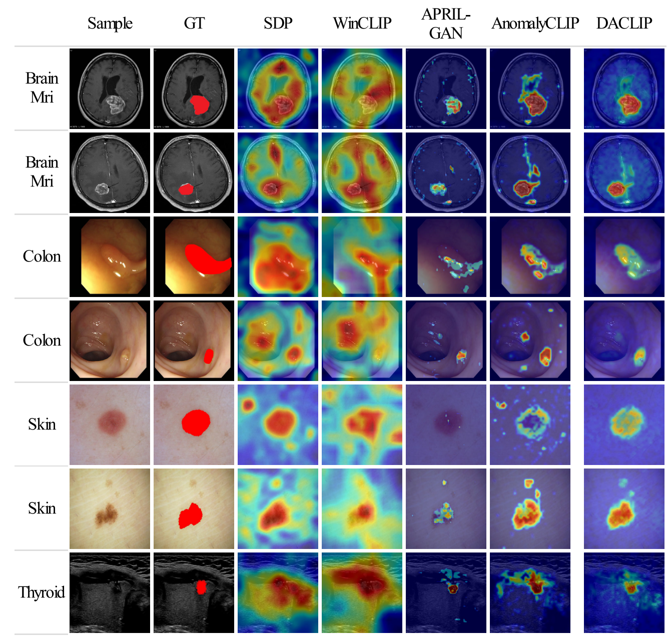

Fig. 6. Qualitative visualization for pixel-level anomaly localization on medical datasets. The leftmost column lists categories, followed by ground truth (GT) masks with annotated anomalies (red). Subsequent columns visualize detection heatmaps from five methods, where warmer hues denote higher confidence in anomaly detection.

methods on the medical domain related datasets. Among the compared methods, DACLIP achieves the best performance on most datasets, demonstrating its effectiveness in ZSAD tasks. Compared to the second-best method (AnomalyCLIP), DACLIP achieves an average improvement of 1.0% in I-Auroc, 1.3% in I-AP and 0.3% in I-F1 on image-level. On pixel-level, DACLIP achieves an average improvement of 5.5% in P-Auroc, 17.5% in P-AP, and 11.3% in P-F1 compared to the AnomalyCLIP. Compared to APRIL-GAN, DACLIP achieves an average improvement of 3.5% in I-Auroc, 2.8% in I-AP and 3.0% in I-F1 on image-level. On the pixel-level, DACLIP demonstrates significant improvements compared to APRIL-GAN, with an average improvement of 9.4% in P-Auroc, 22.8% in P-AP, and 14.9% in P-F1. DACLIP also outperforms SDP, achieving an average improvement of 14.2% in P-Auroc, 30.8% in P-AP and 22.4% in P-F1 on pixel-level. On the other hand, DACLIP achieves an average improvement of 5.2% in I-Auroc, 3.4% in I-AP and 6.2% in I-F1 on image-level compared to SDP. DACLIP achieves better performance than WinCLIP on both image-level and pixel-level metrics. The results indicate that DACLIP is capable of effectively learning domain-aware knowledge and adapting to the target domain, leading to improved performance in ZSAD tasks.

Fig. 6 presents a systematic comparison of anomaly localization performance across five state-of-the-art methods on diverse medical imaging modalities. Notably, DACLIP demonstrates superior localization precision, particularly in challenging cases like subtle colon polyps and heterogeneous skin lesions, where competing methods show either over-detection (WinCLIP) or under-segmentation (APRIL-GAN). The standardized layout enables direct cross-modal performance assessment, revealing method-specific strengths across different imaging characteristics.

TABLE IV

COMPARATIVE RESULTS OF INFERENCE COMPUTATIONAL COSTS. THE INFERENCE TIME IN OUR STUDY IS MEASURED ON A PER-IMAGE BASIS

Metrics	SDP	WinCLIP	APRIL-GAN	AnomalyCLIP	DACLIP
GPU Memory (MiB)	4, 072	3, 814	5, 422	4, 876	5, 546
Inference Time (ms)	25.9	133.3	78.2	112.8	102.0

4) Failure or Similar Qualitative Analysis: We also discuss failure cases where DACLIP shows similar or failed behavior to competing methods, as shown in Fig. 7. DACLIP performs suboptimally in scenarios with extreme occlusions or when anomalies closely mimic normal textures, leading to misclassifications similar or failed to those observed in APRIL-GAN and AnomalyCLIP. In heavily cluttered or occluded scenes the gating network can select suboptimal experts, producing blurred or spurious heatmaps. These analyses underscore the need for further refinement in handling complex visual contexts and subtle anomaly patterns.

5) Inference Computational Cost: All timings were measured on a single NVIDIA RTX 3090 with PyTorch 2.0.0, input resolution 518×518 and batch size 1. As reported in Table IV, DACLIP requires 5,546 MiB of GPU memory and 102.0 ms per image. Relative to AnomalyCLIP (4,876 MiB, 112.8 ms), DACLIP uses around 670 MiB more memory while delivering slightly faster inference. Compared with APRIL-GAN, DACLIP has a small memory increase and a higher

TABLE V
ABLATION STUDY OF THE INFLUENCE OF PROMPT LEARNING ON ZSAD PERFORMANCE. THE METRICS ARE REPORTED IN THE FORMAT OF (I-AUROC, I-AP, I-F1) FOR IMAGE-LEVEL AND (P-AUROC, P-AP, P-F1) FOR PIXEL-LEVEL

Text Prompt	Visual Prompt	MVTec-AD Image-level	MVTec-AD Pixel-level	VisA Image-level	VisA Pixel-level	BrainMRI Image-level
✗	✗	(88.8, 94.7, 91.2)	(87.3, 39.1, 42.0)	(82.4, 85.3, 80.4)	(88.8, 20.6, 26.4)	(93.9, 95.3, 91.1)
✗	✓	(89.9, 95.3, 91.3)	(79.3, 33.1, 37.4)	(82.2, 85.1, 79.8)	(95.5, 25.7, 31.6)	(95.5, 96.6, 94.3)
✓	✗	(89.4, 94.9, 91.4)	(86.4, 36.1, 39.3)	(81.2, 84.4, 79.9)	(95.0, 24.3, 30.5)	(94.4, 96.2, 91.6)
✓	✓	(90.7, 95.8, 92.0)	(88.0, 39.1, 42.2)	(85.2, 87.8, 82.4)	(95.6, 26.3, 32.3)	(95.6, 97.1, 92.4)

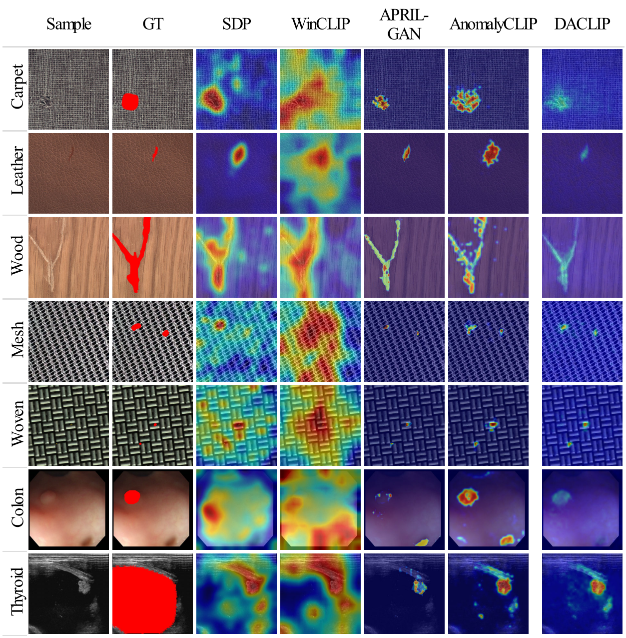

Fig. 7. Failure or similar qualitative results for pixel-level anomaly localization. The leftmost column lists categories, followed by ground truth (GT) masks with annotated anomalies (red). Subsequent columns visualize detection heatmaps from five methods, where warmer hues denote higher confidence in anomaly detection.

latency. The additional cost stems from the DAKA adapters (multiple lightweight experts) and the gating computations; these costs are controlled by sparse Top-K routing and by using compact adapter modules rather than full backbone fine-tuning.

C. Ablation Studies

1) Influence of Prompt Learning: Table V presents the ablation study results on the influence of prompt learning on ZSAD performance. The first row indicates the performance of DACLIP without prompt learning, which serves as the baseline. The second and third rows show the performance of DACLIP with only text prompts or visual prompts, respectively. The last row presents the performance of DACLIP with both text and visual prompts. The results indicate that prompt learning significantly enhances ZSAD performance. The integration of text and visual prompt learning provides robust and flexible adaptation, leading to improved ZSAD performance. With only text or visual prompts, the prediction results remain suboptimal.

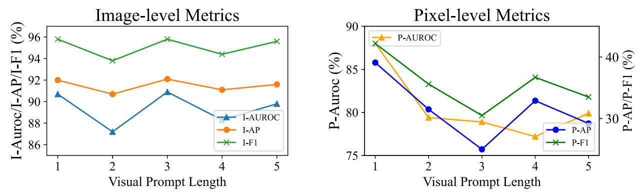

Fig. 8. Experimental results showing the impact of visual prompt length on ZSAD performance when training on VisA and ClinicalDB datasets and evaluating on MVTec-AD.

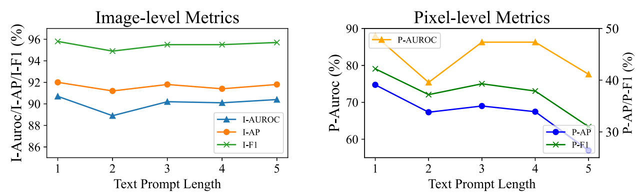

Fig. 9. Experimental results showing the impact of text prompt length on ZSAD performance when training on VisA and ClinicalDB datasets and evaluating on MVTec-AD.

The influence of prompt length on ZSAD performance is analyzed in Fig. 8 and Fig. 9. We vary the visual prompt length from 1 to 5 while keeping the text prompt length fixed at 1 to assess the impact of visual prompt length on ZSAD performance. Similarly, we vary the text prompt length from 1 to 5 while keeping the visual prompt length fixed at 1 to evaluate the effect of text prompt length on ZSAD performance. The results indicate that the length of visual and text prompts significantly impacts ZSAD performance. Longer prompt lengths may lead to overfitting, which degrades performance, especially in the performance of pixel-level anomaly detection.

2) Influence of DAKA Modules: Table VI presents the ablation study results on the influence of DAKA modules on ZSAD performance. The first row indicates the performance of DACLIP without DAKA modules. The second and third rows show the performance of DACLIP with only text DAKA modules or visual DAKA modules, respectively. The last row presents the performance of DACLIP with both text and visual DAKA modules. The findings demonstrate that the DAKA modules play a crucial role in boosting ZSAD performance. Combining text and visual DAKA modules enables effective and adaptable domain-aware learning, resulting in enhanced anomaly detection capabilities.

We also conduct ablation studies to evaluate the impact of integrating DAKA modules at different transformer layers

TABLE VI

ABLATION STUDY OF THE INFLUENCE OF DAKA MODULES ON ZSAD PERFORMANCE. THE METRICS ARE REPORTED IN THE FORMAT OF (I-AUROC, I-AP, I-F1) FOR IMAGE-LEVEL AND (P-AUROC, P-AP, P-F1) FOR PIXEL-LEVEL

Text DAKA	Visual DAKA	MVTec-AD		VisA		BrainMRI
		Image-level	Pixel-level	Image-level	Pixel-level	Image-level
✗	✗	(90.4, 95.7, 91.7)	(86.2, 36.8, 40.9)	(83.5, 86.9, 81.4)	(95.2, 26.3, 31.1)	(93.9, 95.5, 91.3)
✗	✓	(89.6, 95.3, 91.7)	(86.6, 37.9, 41.5)	(83.5, 86.8, 81.2)	(95.3, 25.7, 31.5)	(95.6, 97.1, 92.9)
✓	✗	(90.5, 95.9, 91.4)	(86.1, 38.1, 41.9)	(84.5, 87.3, 81.5)	(95.2, 26.4, 31.6)	(94.2, 95.7, 91.5)
✓	✓	(90.7, 95.8, 92.0)	(88.0, 39.1, 42.2)	(85.2, 87.8, 82.4)	(95.6, 26.3, 32.3)	(95.6, 97.1, 92.4)

TABLE VII

ABLATION STUDY OF THE INFLUENCE OF THE VISUAL TRANSFORMER LAYERS INTEGRATED WITH DAKA MODULES ON ZSAD PERFORMANCE (EVALUATING ON MVTEC-AD). THE METRICS ARE REPORTED IN THE FORMAT OF (I-AUROC, I-AP, I-F1) FOR IMAGE-LEVEL AND (P-AUROC, P-AP, P-F1) FOR PIXEL-LEVEL

Visual Layers	MVTec-AD	
	Image-level	Pixel-level
(1)	(89.5, 95.3, 91.4)	(85.9, 35.9, 40.1)
(1, 2)	(91.3, 96.0, 92.3)	(79.9, 32.7, 37.0)
(1, 2, 3)	(89.2, 95.1, 91.1)	(84.1, 37.2, 41.9)
(2, 3, 4)	(89.7, 95.2, 91.5)	(84.2, 37.0, 41.7)
(1, 2, 3, 4)	(90.7, 95.8, 92.0)	(88.0, 39.1, 42.2)

TABLE VIII

ABLATION STUDY OF THE INFLUENCE OF THE TEXT TRANSFORMER LAYERS INTEGRATED WITH DAKA MODULES ON ZSAD PERFORMANCE (EVALUATING ON MVTEC-AD). THE METRICS ARE REPORTED IN THE FORMAT OF (I-AUROC, I-AP, I-F1) FOR IMAGE-LEVEL AND (P-AUROC, P-AP, P-F1) FOR PIXEL-LEVEL

Text Layers	MVTec-AD	
	Image-level	Pixel-level
(1)	(89.7, 95.5, 91.5)	(86.0, 37.2, 40.6)
(1, 2)	(89.3, 95.0, 91.4)	(85.8, 38.7, 41.5)
(1, 2, 3)	(91.1, 95.5, 91.8)	(82.2, 36.1, 40.2)
(2, 3, 4)	(89.4, 94.9, 91.4)	(76.8, 29.9, 33.0)
(1, 2, 3, 4)	(90.7, 95.8, 92.0)	(88.0, 39.1, 42.2)

within the visual and text encoders of CLIP. Table VII presents the results of integrating DAKA modules at various combinations of visual transformer layers. The findings indicate that incorporating DAKA modules across multiple layers enhances ZSAD performance, with the best results achieved when modules are added to all four selected layers. This multi-layer integration allows the model to capture a richer domain-aware features, improving both image-level and pixel-level anomaly detection. Table VIII shows the results of integrating DAKA modules at different text transformer layers. Similar to the visual encoder, the best performance is observed when DAKA modules are integrated across all four selected text layers. This comprehensive integration enables the model to effectively learn domain-specific textual representations, further boosting ZSAD capabilities.

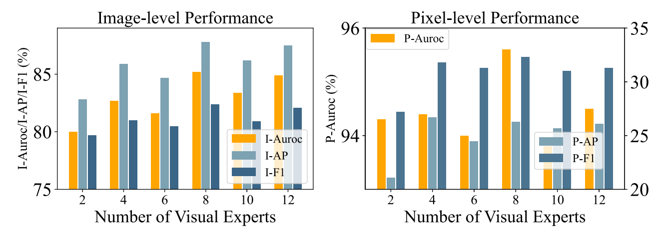

Fig. 10. Experimental results of visual expert number evaluating on VisA.

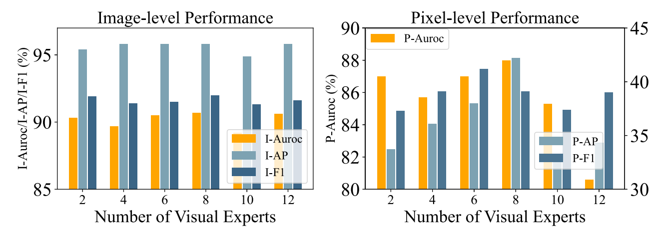

Fig. 11. Experimental results of visual expert number evaluating on MVTec-AD.

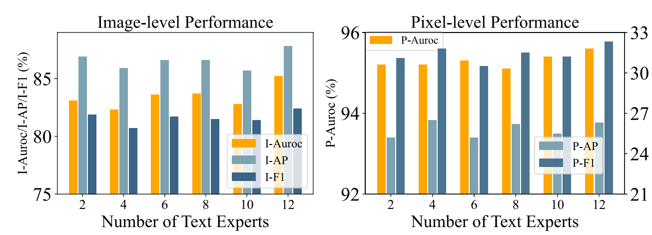

Fig. 12. Experimental results of text expert number evaluating on VisA.

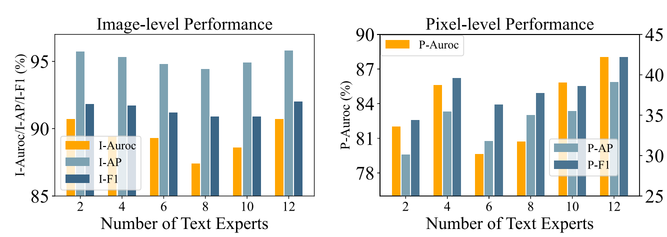

Fig. 13. Experimental results of text expert number evaluating on MVTec-AD.

3) Analysis on the Number of Experts: As shown in Fig. 10 and Fig. 11, we analyze the influence of the number of visual experts on ZSAD performance. We also analyze the influence of the number of text experts on ZSAD performance in Fig. 12 and Fig. 13. The number of text experts is set to 12, while performing the analysis on the number of visual experts. The

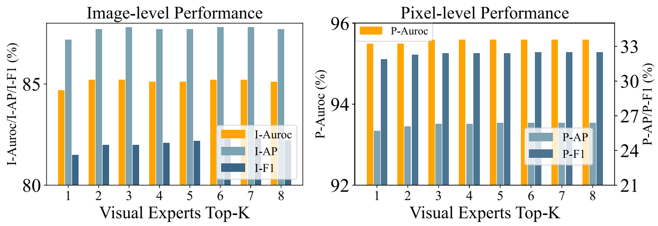

Fig. 14. Experimental results of visual experts Top-K evaluating on VisA.

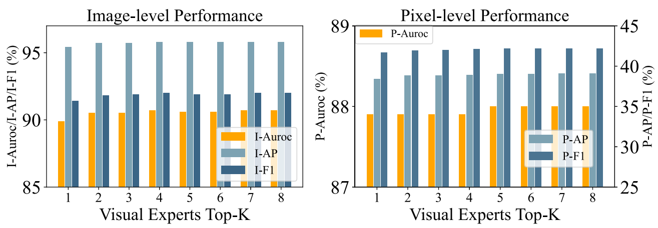

Fig. 15. Experimental results of visual experts Top-K evaluating on MVTec-AD.

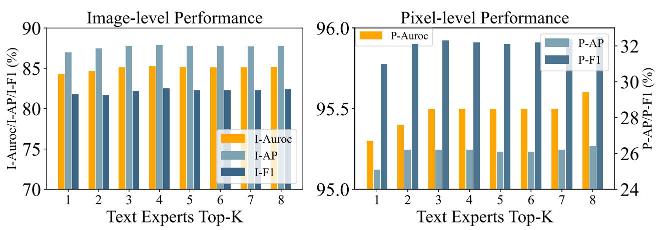

Fig. 16. Experimental results of text experts Top-K evaluating on VisA.

number of visual experts is set to 8, while performing the analysis on the number of text experts. The results indicate that the number of visual and text experts significantly impacts ZSAD performance. The performance improves as the number of experts increases, but the improvement diminishes after reaching a certain threshold. The optimal number of visual experts is 8, which achieves the best performance on ZSAD tasks. The optimal number of text experts is 12, which achieves the best performance on ZSAD tasks. Insufficient experts limit the model's capacity to specialize, degrading performance; excessive experts risk overfitting and raise computational cost. The results suggest that the number of experts should be carefully selected to balance performance and computational efficiency. We set the number of visual experts to 8 and the number of text experts to 12 in our experiments.

4) Analysis on the Number of Top-K: As shown in Fig. 14 and Fig. 15, we analyze the influence of the number of Top-K for visual experts on ZSAD performance. We present the analysis results of the number of Top-K for text experts on ZSAD performance in Fig. 16 and Fig. 17. When analyzing different Top-K values for visual experts, the Top-K for text experts is held constant; conversely, when studying Top-K for text experts, the visual Top-K is fixed. The results indicate that the number of Top-K of experts has a significant impact on ZSAD performance. The performance improves when selecting a suitable Top-K number of experts, but the improvement diminishes after reaching a certain threshold. The text Top-K

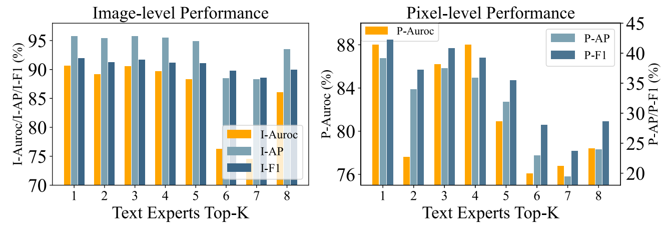

Fig. 17. Experimental results of text experts Top-K on MVTec-AD.

TABLE IX
ABLATION STUDY OF THE INFLUENCE OF THE RESOLUTION OF INPUT IMAGES ON ZSAD PERFORMANCE (EVALUATING ON MVTEC-AD). THE METRICS ARE REPORTED IN THE FORMAT OF (I-AUROC, I-AP, I-F1) FOR IMAGE-LEVEL AND (P-AUROC, P-AP, P-F1) FOR PIXEL-LEVEL

Resolution	MVTec-AD	
	Image-level	Pixel-level
(224×224)	(87.4, 94.3, 89.8)	(89.1, 33.9, 37.7)
(336×336)	(89.5, 95.2, 90.4)	(89.2, 38.7, 41.8)
(518×518)	(90.7, 95.8, 92.0)	(88.0, 39.1, 42.2)

has a greater influence on ZSAD performance compared to visual Top-K. This may be because text experts capture higher-level semantic information, making their selection more critical for accurate anomaly detection. Moreover, the optimal Top-K for text experts differs significantly across datasets: Top-K= 1 for MVTec-AD versus Top-K= 8 for VisA. This discrepancy reflects distinct dataset characteristics. Anomalies of MVTec-AD often share common semantic patterns where a single text expert can effectively learn a universal "abnormality" concept that generalizes across diverse categories. In contrast, VisA contains anomalies with fine-grained, category-specific anomaly semantics, which require multiple specialized text experts to capture effectively. These results suggest that optimal Top-K should be tuned based on dataset characteristics.

5) Influence of Input Image Resolution: Table IX presents the ablation study results on the influence of input image resolution on ZSAD performance. We evaluate the performance of DACLIP at three different input resolutions: 224×224, 336×336, and 518×518. Although P-Auroc shows a slight decrease when the resolution is increased, the overall trend indicates that higher resolution improves the model's ability to capture the fine-grained details essential for accurate anomaly detection. The overall results indicate that higher input image resolutions lead to improved ZSAD performance.

6) Analysis of the Different Loss Function Weights: As shown in Fig. 18, we analyze the influence of different loss function weights on ZSAD performance. We set the global loss weight $\mathcal{L}_{global}$ to a fixed value and systematically vary the local loss weight $\mathcal{L}_{local}$ from 0.1 to 1.0 to evaluate the sensitivity of the model to this hyperparameter. The results indicate that the weight of the loss function does not significantly influence the performance of ZSAD. This robustness to weight adjustments can be attributed to the complementary nature of

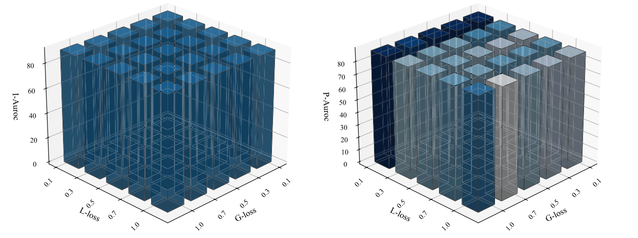

Fig. 18. Experimental results showing the impact of different loss function weights when training on VisA and ClinicalDB datasets and evaluating on MVTec-AD. The deeper the color, the higher the performance.

the two loss functions: the global loss optimizes image-level anomaly classification, while the local loss refines pixel-level localization. Since these objectives target different aspects of the model's learning process, variations in their relative weighting have limited impact on overall performance.

V. Conclusion

In this paper, we propose a novel zero-shot anomaly detection method called DACLIP, which leverages the power of CLIP and introduces DAKA to enhance the model's adaptability to unseen domains. DACLIP effectively learns domain-aware knowledge and adapts to the target domain, leading to improved performance in ZSAD tasks. DACLIP employs a domain-aware gating mechanism to select the most relevant experts for each input, allowing more efficient and effective domain adaptation. DACLIP also incorporates prompt learning to enhance the model's ability to understand and represent the target domain. We utilize both text and visual prompts to guide the model in learning domain-aware features, enabling it to better capture the characteristics of the target domain. Additionally, we introduce a projection layer to align the dimensions of patch embeddings with text embeddings, enabling effective anomaly detection. We conducted extensive experiments on various datasets from both industrial and medical domains, demonstrating that DACLIP outperforms state-of-the-art methods in terms of image-level and pixel-level anomaly detection performance.

1) Limitations and Future Work: While DACLIP improves zero-shot anomaly detection across diverse domains, several limitations and broader impacts deserve attention. Common failure modes include very small or low-contrast anomalies, texture-only defects that induce minimal semantic shift, and domain shifts not covered by our auxiliary training data; these issues can degrade localization accuracy and produce miscalibrated confidence scores. In safety-critical applications (e.g., medical diagnosis or industrial inspection), false negatives may cause harm and false positives can lead to unnecessary interventions and cost. Therefore, model outputs should be treated as decision support rather than sole determinations and must be validated in the target operational context. Future work will focus on explainability for routing decisions, formal robustness evaluations under distribution shift, multi-scale and higher-resolution adaptations, and uncertainty estimation and human-in-the-loop strategies to mitigate deployment risks.

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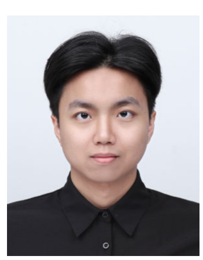

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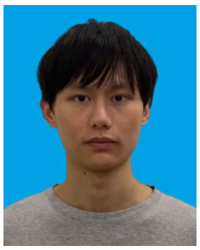

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